



Test engineers are turning to a software-defined approach to instrumentation so that they can quickly customise their equipment to specific application needs and integrate testing directly into the design process. Not only does this approach provide flexibility, it also gives test engineers the ability to take advantage of the performance provided by the latest PC, CPU and bus technologies” — Satish Mohanram, technical marketing manager, National Instruments - India

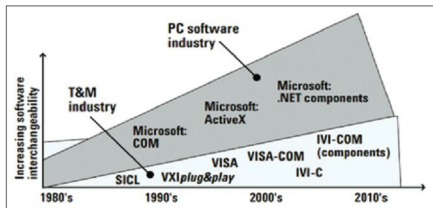


Image Courtesy: Agilent's application note 'Test-System Development Guide' AN 1465-3

certainly out of reusing application libraries and changing vendor-specific hardware.

Dhillon explains, "The end user can now focus on the requirement rather than being limited by a single vendor's hardware. He can mix and match between hardware from various vendors to choose the hardware that exactly fits his requirement and yet be able to integrate them in one test system with a single application controlling and automating the entire setup. This is quite different from the situation about two decades ago when there was no unifying platform and vendors strug-

gled to get a newer variant of their own hardware to work with their software framework. Of course, inter-vendor operability was a far cry at that time."

PXI seems to be the uniting standard for most T&M equipment.

Hire explains: "The PXI standard moves automation forward in a number of ways. First, with a separate PC controller and high-speed Ethernet, test speed improves. The ability to upgrade the PC processor to take advantage of faster models also delivers valuable and cost-effective improvements. In comparison, traditional test equipment were stuck with the original processor with

which these were designed and an upgrade usually meant replacing the entire instrument. When a factory is producing several million products a month, even a 2-3-second speed gain from improved processing power makes a large cumulative difference."

"Second, PXI separates the test hardware from software drivers. The benefit of this is that the same modules can be used for a variety of different applications. Manufacturers can run different programs on the same hardware and still have the advantage of optimising the test performance for the specific end product. As products change, new software can be developed in parallel—or if a production line needs to make several different variants of the product, multiple software applications can be used to quickly reconfigure a line to meet changing customer needs," he adds.

These benefits have enabled a huge number of equipment compatible with PXI platform. "Being an open PC-based platform, PXI offers a wide variety of real-time processor options including several high-performance multi-core processors. With more than 1200 products from more than 70 vendors, it is the platform of choice for thousands of companies worldwide," adds Mohanram.

Coming up next...

Test automation has progressed by leaps and bounds over the years, giving users the option to choose from wireless portable and real-time automation solutions. Networking capabilities offered by T&M solutions are another factor helping in easier reporting. So where is test and measurement headed now?

"Test and measurement is much more aligned with mainstream technology than ever before. We can be sure that it will go hand in hand with the newest technologies and even play a part in defining them. Maybe we can expect voice-input-controlled instruments and 3D signal visualisation tools in the time to come," answers Dhillon. ●

The author is a tech correspondent at EFY



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